

ALEKHYA MULIKI

PHONE: +1 (769) 280-0437 | EMAIL: alekhyamuliki1997@gmail.com | www.linkedin.com/in/alekhya-muliki

SUMMARY

Dedicated Public Health professional with a Master of Public Health (MPH) in Epidemiology and Biostatistics and a Doctor of Pharmacy (PharmD) degree. Proficient in applying statistical software tools such as SAS, SPSS, and R for advanced data analysis and interpretation. Experienced in public health research, biostatistics, data management, and health data analytics. Demonstrated expertise in developing research proposals and delivering scientific presentations. Skilled in utilizing REDCap and other reporting systems for secure data collection and management. Committed to evidence-based practices and improving population health outcomes through rigorous epidemiological methods.

EDUCATION

The University of Southern Mississippi

Master of Public Health - Epidemiology and Biostatistics. GPA 3.89

Hattiesburg, MS

Jan 2024 – May 2025

Jawaharlal Nehru Technological University Hyderabad

Doctor of Pharmacy (PharmD). Grade A+

Telangana, India

Sep 2015– Aug 2021

SKILLS

- Technical Skills: Proficient in SAS® 9.4, SPSS, Python, SQL, R (including R Markdown), REDCAP, QUALTRICS, JASP, Power BI, Tableau, Epi Info and epiR. Also Proficient in Microsoft Office Suite, including Excel, PowerPoint, Word, Access, and Teams, for data organization, reporting, presentations, and collaborative project management.
- Epidemiology: Experienced in study designs and outbreak investigations to identify disease patterns and risk factors.
- Data Management & Analysis: Skilled in data cleaning, transformation, visualization, handling missing values, validation, and applying statistical methods including regression, survival analysis, and hypothesis testing.
- Reporting & Clinical Research: Adept at creating tables and graphs for epidemiological reports, designing Case Report Forms, and ensuring Good Clinical Practice compliance.

EXPERIENCE

Health Data Analyst Intern

Jan 2025 – May 2025

Gulf South Center for Community Engaged Health Research and Innovation, Hattiesburg, MS, USA 39401.

- Led the public health project “Impact of Telehealth and Broadband Access on Accessibility and Quality of Healthcare Services in Mississippi: A County Level Study” applying advanced statistical data analysis using SAS, R, JASP, SPSS and MS Excel.
- Worked on DHQ3 Data Reentry by comparing with REDCAP Data file and created surveys using Qualtrics.
- Conducted in-depth analysis of the Effect of Dietary intake on Sleep Quality using statistical software SAS, R Software and MS Excel.
- Worked on ASPIRE (Advancing Social and Environmental Equity through Plastics Research: Education, Innovation, and Inclusion) Project which is a collaborative study between School of Health Professions and Polymer Science Department.
- Worked on Nutrition 360 Project to establish essential components to address diet quality in healthcare settings and examine what value an integrated service model delivery has for the primary prevention of cardiometabolic diseases with an initial focus on dietary behaviors.
- Utilized public health surveillance systems, including the BRFSS, NHANES datasets, to collect datasets for analysis.
- Performed biostatistical tests such as Correlation, Chi-square, ANOVA, Linear and Logistic Regression to analyze relationships between variables.
- Developed abstracts, poster presentations, and flyers to disseminate research findings to stakeholders, ensuring accessibility and actionable insights.
- Managed and optimized the REDCAP Survey Library, improving data entry accuracy and data collection efficiency for ongoing research projects.

Student Worker

Jan 2025 - May 2025

The University of Southern Mississippi, School of Health Professions, Hattiesburg, MS, USA 39401.

- Worked on the SRH Project titled “Impact of Telehealth and Broadband Access on Accessibility and Quality of Healthcare Services in Mississippi : A County Level Study”
- Conducted statistical analysis using R and Excel, including descriptive statistics, correlation analysis, chi-square tests, and Multiple logistic regression to support faculty-led healthcare accessibility studies focused on broadband access and telehealth utilization.
- Analyzed secondary data from Mississippi State Department of Health, CDC, and Surgo Ventures to assess healthcare accessibility disparities.
- Presented findings on health equity, digital infrastructure, and policy implications for improving telehealth services.
- Assisted in data collection, cleaning, literature reviews and management for public health research projects.
- Prepared reports, presentations, and visualizations on telehealth adoption and digital health disparities.
- Engaged with community stakeholders and healthcare professionals to discuss telehealth implementation challenges.

Fortrea Clinical Development Pvt Ltd -Bangalore,India.

- Developed and programmed SDTM (Study Data Tabulation Model) datasets for clinical trials.
- Created and analyzed ADaM (Analysis Data Model) datasets to support statistical analysis.
- Generated TLFs (Tables, Listings, and Figures) for clinical study reports.
- Conducted statistical research and analysis on clinical trial data.
- Compiled and interpreted statistical records to support research projects.
- Validated final statistical SDTM datasets, ensuring data accuracy and integrity.
- Contributed to the statistical programming aspects of drug development projects.
- Applied knowledge of clinical trial data structures and standards.

Clinical SAS Intern Programmer**Sep 2021 – Apr 2022**

Aculore Statistics Pvt Ltd. - Hyderabad, Telangana, India

- Good knowledge on Functions, Formats, Informats and Conditional statements.
- Expertise in using Drop, Keep, Rename options in data steps.
- Good knowledge in using Character Functions, Numeric functions and Case functions.
- Proficient in Base/SAS, SAS/Macro and SAS/SQL.
- Basic knowledge of CDISC standards like SDTM (3.2) Domains.
- Familiar with SAS procedures, do loops (do while, do until).
- Good basics on SAS/macro programming, including macro debugging techniques and utilizing macros with keyword and positional parameters.

Clinical Research Coordinator Intern**Jun 2020 – Aug 2021**

Gandhi Medical College and Hospital (GMC)Hospitals - Hyderabad, India

- Coordinated case collection and management of clinical data related to drug-drug interactions, adverse drug reactions (ADRs), and drug information queries to support clinical trial objectives and decision-making.
- Assisted in organizing and facilitating ward rounds by collaborating with physicians, pharmacists, and healthcare teams to ensure accurate data capture, while supporting the identification, documentation, and reporting of medication errors and ADRs in compliance with Good Clinical Practice (GCP) and pharmacovigilance regulatory requirements.
- Performed phlebotomy on hospitalized patients to collect blood specimens, followed by laboratory testing including complete blood picture (CBP), LFT, HbA1c, vitamin D levels, and other condition-specific diagnostics. Conducted statistical analysis using SAS, applying biostatistical methods to calculate prevalence, incidence, and statistical significance (p-values).
- Contributed to a prospective observational cohort clinical trial on vitamin D deficiency and its association with respiratory tract infections in pediatric patients, assisting with data coordination and documentation following GCP standards.

PUBLICATIONS

- "Vitamin D Deficiency and its Association with Respiratory Tract Infections in Pediatrics: Life Sciences-Pharma" – Published in *IJLPR International Journal of Life Sciences and Pharma Research*.
- "A Case Report on Bullous Fixed Drug Eruption" – Published in *International Journal of Pharmaceutical Research*, Volume 13, Issue 2, June 2021.
- "A Case Report on Thyrotoxicosis-Induced Myopathy" – Published in *Pensee International Journal*, Volume 51, Issue 04, April 2021.
- "A Review on Medication Errors in a Tertiary Care Hospital" – Published in *Pensee International Journal*, Volume 51, Issue 5, May 2021.

CERTIFICATIONS

- SAS Programming Fundamentals Using SAS® 9.4
- Statistics/Data Analysis in SPSS: Inferential Statistics
- R Programming Complete Certification Training
- CITI Human Subjects Research
- Good Clinical Practice for Clinical Research Professionals / NIDA Clinical Trials Network
- Decision-making and Risk Communication in Emergency Preparedness.
- Introduction to the Clinical Research Principles and Practices by National Institutes of Health (NIH)

POSTERS AND PRESENTATIONS

- Awarded “**MPH RESEARCH AWARD**” at school of health professionals, University of southern Mississippi, Hattiesburg, Ms,2024.
- Conducted awareness sessions on “**Vitamin D Deficiency and Its Association with Respiratory Tract Infections in Pediatrics**” Hyderabad, India.
- Statistical presentation on “**Impact of Telehealth and Broadband Access on Accessibility and Quality of Healthcare Services in Mississippi: A County Level Study**” / SAS, R, Excel and SPSS for data analysis and visualization.

RESEARCH PROJECTS

- Analysis report on The Effect of dietary intake on Sleep Quality/ **SAS**.
- Mississippi Department of Health Dataset 2022 Data Cleaning Project / **R Programming and SAS**.
- Impact of Telehealth and Broadband Access on Accessibility and Quality of Healthcare Services in Mississippi: A County Level Study”/ **R Programming, SPSS and SAS**.
- Association between the Effect of Stress on Physical Activity in USA/ **Systematic reviews**.